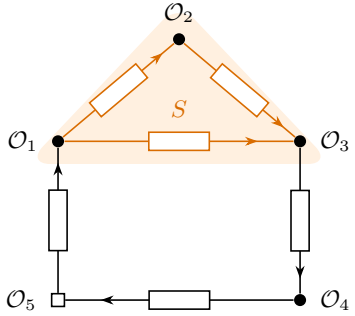
(a) Γ on the entries, $k=5$, $|\Gamma_1| = 2k-4$

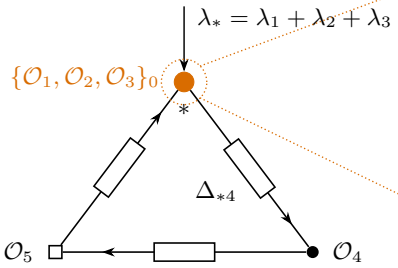


grounded node $k=5$

$$\mathrm{sgn}(\pi_S)\, W_\Gamma = \underbrace{\Delta_{12}\Delta_{13}\Delta_{23}}_{W_{\Gamma[S]}}\underbrace{\Delta_{34}\Delta_{45}\Delta_{51}}_{W_{\Gamma(S)}}(\mathcal{O}_1\cdots\mathcal{O}_5)$$

$$\mathrm{sgn}(\pi_S) = +1$$

(b) The outer bracket $\{P^{(S)}\}_0$

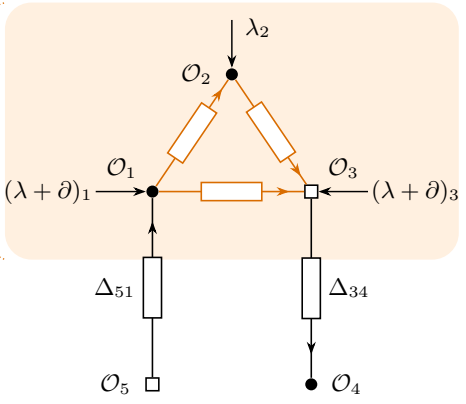


$$\Delta_{*4} = \Delta_{14} + \Delta_{24} + \Delta_{34}$$

$$P^{(S)} = \big(\{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3\}_0,\ \mathcal{O}_4,\ \mathcal{O}_5\big)$$

$$i_{|S|} = 3 < j_1 = 4, \quad p = 0, \quad \tau_S = (-1)^{q_S} = +1$$

(c) The island bracket $\{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3\}_0$



$$\lambda_{3,+\dot a} \longmapsto +\frac{\partial}{\partial z_{34}^{+\dot a}}, \qquad \lambda_{1,+\dot a} \longmapsto -\frac{\partial}{\partial z_{51}^{+\dot a}}$$

$$(\lambda+\partial)_{3,+\dot a} \ \to\ -(\lambda+\partial)_{1,+\dot a} - (\lambda+\partial)_{2,+\dot a}$$